Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
And
KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Rebuttal Testimony

of

The Capital Structure Panel

Case 23-G-0225
Case 23-G-0226

Dated: September 22, 2023

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Rebuttal Testimony of the Capital Structure Panel

I. Introduction

Q. Please introduce the members of the Capital Structure Panel.

A. The Panel consists of Christopher Porter and Keith Magee.

Q. Is this the same panel that previously submitted testimony in these proceedings?

A. Yes. The defined terms in the Panel's direct testimony have the same definitions here unless otherwise indicated.

II. Purpose of Testimony

Q. What is the purpose of your rebuttal testimony?

A. The purpose of our rebuttal testimony is to address certain cost of capital and capital structure recommendations made by Department of Public Service Staff ("Staff") witness Kwaku Duah and New York City ("NYC") witness Michele Chait. First, on the issues of equity ratio and ring fencing, Mr. Duah is aligned with the Companies. Staff Witness Duah (at 19) "agree[s] with the Companies that they have implemented strong ring-fencing measures that have been acknowledged by both Moody's and S&P." Mr. Duah also "recommend[s] the use of a 48.0 percent common equity ratio for ratemaking purposes because [he] conclude[s] that such a common equity ratio is reasonable." Despite agreement with the Companies on these major issues, there still exist several areas of disagreement. Specifically, we will address and rebut Staff Witness Duah's: (1) proposed cost rate of debt and projected cost rates; (2) financial integrity analysis; (3) recommendation to forgo a long-term debt cost reconciliation mechanism; and (4)

proposed capital structure for the Service Company. We will also address NYC Witness Chait's proposal to reject the Companies' proposed long-term debt reconciliation mechanism.

Q. Do other witnesses address aspects of Staff Witness Duah's recommendation concerning the Companies' cost of capital?

A. Yes. Company Witness Josh Nowak's rebuttal testimony addresses Staff Witness Duah's recommendation concerning the Companies' requested return on equity ("ROE"). Taken together, our rebuttal and Mr. Nowak's rebuttal support a finding that the Companies' proposed capital structure and overall rate of return should be used for ratemaking purposes in these proceedings.

Q. Does the Panel sponsor any exhibits?

A. Yes. The Panel sponsors Exhibit___(CSP-1R), Schedule 1, which sets forth the Companies' updated cost of long-term debt, and Schedule 2, which sets forth the Companies' updated weighted cost of capital for the Rate Year and Data Years. The exhibit and schedules were prepared and compiled under our direction and supervision.

III. <u>Debt Costs and Debt Cost Projections</u>

Q. Does Staff Witness Duah accept the Companies' historic cost of debt and credit spread analysis?

A. Yes, he does.

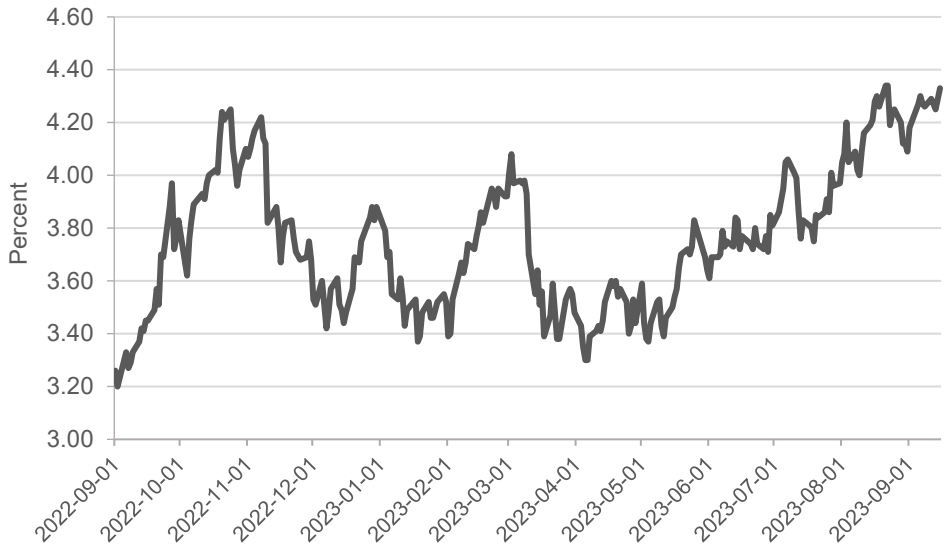
Q. Does he agree with the Companies' projected cost of long-term debt?

A. No. While he accepts the Panel's proposed credit spreads, Staff Witness Duah does not agree with the Panel's use of forward Treasury yields to determine the interest rate of the planned Rate Year debt issuance. Instead, he recommends (at 54) the use of current U.S. Treasury yields plus the Companies' calculated credit spreads to estimate interest rates for the Companies' projected new debt issuances. Mr. Duah claims that "current interest rates are the best predictors of future interest rates." He further argues that "this approach is more reliable for predicting future interest rates, particularly for a one-year rate plan in these rate proceedings."

Q. Do you agree?

A. No. The Federal Reserve Economic Data on recent 10-year U.S. Treasury bond yields in Figure 1, below, illustrates that current yields at any given point in time are not the best predictors of future interest rates. We recognize the Federal Reserve has been raising the federal funds rate (*i.e.*, the inter-bank lending rate) and normalizing its balance sheet to address inflation. However, the specific path of the Federal Reserve's monetary policy is uncertain and will respond to changing market conditions. We do not believe setting future rates based on a point in time during the Federal Reserve's monetary tightening cycle would be reasonable. Figure 1 demonstrates that a point-in-time estimate using recent market conditions is not the appropriate estimate of Treasury yields over the period that the rates determined in this proceeding will be in effect.

Figure 1: Ten-Year Treasury Bond Yield (Sept 1, 2022 – Sept 15, 2023)*



*Federal Reserve Economic Data *available at* https://fred.stlouisfed.org/series/DGS10

In addition, the implied 10-year and 30-year Treasury yields used by the Companies are derived from the current Treasury yield curve. Thus, they represent market-implied future yields of 10-year and 30-year Treasuries and therefore should reflect market participants' views on interest rates. For these reasons, we strongly disagree with Staff Witness Duah's recommendation to assume current yields for future debt issuances. We note, however, that if the Companies' proposed long-term debt true-up mechanism is approved the interest expense on future debt issuances will be reconciled to actual incurred interest expense and, therefore, the difference between Staff Witness Duah's and the Companies' forecast rate assumptions will be moot.

Q. Are there any updates to the Companies' debt cost rates?

A. Yes, in Exhibit ___ (CSP-1R), Schedule 1 we have updated the debt cost rates for both KEDNY and KEDLI to reflect movements in forward Treasury yields and credit spreads,

and for KEDNY we have reflected the known and measurable costs associated with the $400 million 10-year bond recently issued on September 15, 2023. This bond issuance replaces the planned $500 million bond, which was assumed to be issued in July 2023 in the Panel's direct testimony. There are no other changes to the amounts or issuance dates for other future debt issuances. The coupon on KEDNY's recent issuance of 6.388 percent was set based on a credit spread of 2.12 percent over the 10-year Treasury yield of 4.268 percent. This is a slightly higher coupon than the 6.229 percent rate assumed in the Companies' supplemental response to Information Request ("IR") No. DPS-532 when the Companies last refreshed their future debt issuance rates (included in Staff Witness Duah's Exhibit___(KXD-1)), which was based on a credit spread of 2.20 percent over a forward Treasury yield of 4.029 percent. KEDNY's recent bond issue indicates the current credit spread required by bond investors to invest in new KEDNY 10-year debt. As this credit spread is 0.08 percent lower than that assumed for 10-year KEDNY debt in the response to IR No. DPS-532, we have reduced the credit spread assumption for all future KEDNY and KEDLI issuances by 0.08 percent. The corresponding credit spreads are then added to forward Treasury yields, sourced from Bloomberg as of September 12, 2023 to produce the coupon rates for each future issuance.

IV. <u>Proposed Long-Term Debt True-Up Mechanism</u>

Q. Does Staff Witness Duah agree with the Companies' proposed long-term debt true-up mechanism?

A. No. Staff Witness Duah (at 61) does not agree with the Companies' proposal to implement a long-term debt true-up mechanism. He argues, "[g]iven the current stability in the capital

markets, the Companies should be able to access the markets with reasonable certainty during the Rate Year without the need for a true-up." As shown by KEDNY's recent bond issuance, however, the risk even exists in a one-year case. The 10-year Treasury yield was 4.268 percent at the time of pricing, around 0.69 percent (69 basis points) higher than six months previously. Interest rates, as illustrated in Figure 1, above, can be volatile and unpredictable. In 2022, alone, the Federal Reserve raised interest rates seven times, going from a Federal Funds rate range of 0.25 to 0.50 percent to a range of 4.25 to 4.50 percent. The Federal Reserve then raised rates by an additional 100 basis points in 2023, to a Federal Funds rate range of 5.25 to 5.50 percent. At this point, although it appears that the Federal Reserve may be pausing rate increases, rates have not been at 5.5 percent since early 2001. The 10-year Treasury yield is also at its highest level since 2007. The debt cost tracker proposed by the Companies will protect both the Companies and their customers. For example, if long-term Treasury yields decline compared to the forward Treasury yields that we have used to set future debt cost rates, the true-up mechanism will ensure that customers will not pay higher rates than necessary.

Q. What is your response to NYC Witness Chait's contention that there is no evidence that recent interest rate volatility will impact the future cost of new long-term debt?

A. As discussed above, there has been significant interest rate volatility over the last several years and there is no reason to believe the likelihood of interest rate changes has declined. There continues to be significant uncertainty about factors such inflation, Federal Reserve market policy, and bond market volatility that affect interest rates. 10-year and 30-year

Treasury yields could move in either direction, and neither customers nor the Companies would be disadvantaged by a reconciling mechanism for long-term debt costs.

V. Financial Integrity

Q. Do you agree with Staff Witness Duah's assertion (at 36) that his recommended common equity ratio and ROE are generally consistent with the performance required by Moody's and S&P to maintain the Companies' current credit ratings?

A. Not entirely. Beginning with the analysis of KEDNY's S&P rating (at 44) and in Exhibit __ (KXD-12), Staff Witness Duah indicates that his analysis shows an implied S&P rating of "BBB-". This is not consistent with KEDNY's existing BBB+ issuer credit rating. S&P currently assigns KEDNY a stand-alone credit profile ("SACP") of 'bbb' and an actual issuer credit rating of BBB+. The SACP is S&P's opinion of creditworthiness in the absence of extraordinary intervention or support from a parent, affiliate, or related government or third-party entity. The actual rating assigned is one notch higher at BBB+ due to KEDNY's status as a core subsidiary of National Grid North America ("NGNA"), which, in turn, is a core subsidiary of National Grid plc ("NG"), both of which have a "BBB+" issuer credit rating. Mr. Duah's implied rating of "BBB-" would only result in a BBB+ assigned rating if NGNA and NG continue to maintain BBB+ ratings and if S&P does not alter its approach under its group rating methodology. We believe that it is most appropriate for KEDNY to maintain a stand-alone rating profile in line with its actual BBB+ rating, as otherwise a change in S&P's approach could lead to a downgrade, which would increase future debt costs. The credit ratings of NGNA and NG are also a function of the credit ratings of their constituent parts and therefore it is important for individual

operating company subsidiaries, like KEDNY, to maintain supportive stand-alone rating profiles. In addition, we note that Staff Witness Duah's BBB- implied rating is not in line with the "Generic Financing Proceeding" (Case 91-M-0509), as mentioned in the Panel's original testimony, which proposed that the "A" rating should continue to be the long-term target for utilities.

Q. Do you agree with Staff Witness Duah's assessment of KEDLI's S&P rating?

A. Yes. Although we cannot speak for the rating agencies, we believe his assessment is fair. In its latest report in February 2023, S&P assigns a 'bbb+' stand-alone credit profile and a BBB+ issuer credit rating to KEDLI and indicate that they "expect the company's funds from operations (FFO) to debt to average 13%-16% through our forecast period, evaluated under our medial volatility financial benchmark table." Staff Witness Duah's calculation of S&P's FFO/debt for the Rate Year is 15.04 percent, which is within this range (at 47).

Q. What implied Moody's ratings did Staff Witness Duah calculate for KEDNY and KEDLI?

A. Staff Witness Duah indicates (at 43) that his recommendations for the equity ratio and ROE, alongside recommendations from the Staff Revenue Requirements and Staff Gas Infrastructure and Operations panels, produced credit metrics consistent with a Baa1 rating for KEDNY and a A3 rating for KEDLI.

Q. Do you agree with this conclusion?

A. No. We believe the Moody's Cash Flow from Operations ("CFO") pre-WC/debt metric provides the best indication of the credit ratings the rating agency will likely assign to the Companies. In its latest published reports on KEDNY and KEDLI (Exhibits ___ (KXD-5) and ___ (KXD-6), respectively), Moody's sets out factors that could lead to a rating upgrade or downgrade. Among other qualitative factors Moody's states that KEDNY's rating could be upgraded "if its ratio of CFO pre-WC/debt were to be at least 15% on a sustainable basis," and it could be downgraded if the ratio was "persistently below 13%" and there was "a deterioration of the consolidated credit quality of the National Grid group." Similarly for KEDLI, alongside other qualitative factors, Moody's indicates the utility's rating could be upgraded if the ratio "were to be at least 19% on a sustainable basis under its next rate plan" and that "downward rating pressure could arise" if the metric "was expected to fall persistently below 15%, or if state energy policy resulted in a deterioration in either cash flow predictability; or ability or timeliness of cost recovery." Therefore, KEDNY and KEDLI ought to maintain Moody's CFO pre-WC/debt ratios at least above 13 percent and 15 percent, respectively, to avoid the risk of a rating downgrade.

Q. What CFO pre-WC/debt metrics did Staff Witness Duah calculate for the Rate Year for the Companies, and did you find any inconsistencies in the calculations?

A. For KEDLI Staff Witness Duah calculated a metric of 14.65 percent and for KEDNY a metric of 14.95 percent. However, in the calculation of the line "Moody's Unadjusted CFO pre-WC" for KEDNY there is a formulaic error. "Forecasted changes in Working Capital" was subtracted from Net Income, however, this is not a necessary step and was not done

for KEDLI. Correcting this reduces KEDNY's CFO pre-WC/debt by $17.808 million and produces a CFO pre-WC/debt metric of 14.52 percent.

Q. What ratings are implied by KEDNY's updated CFO pre-WC/debt metric of 14.52 percent and KEDLI's metric of 14.65 percent?

A. As mentioned previously, Moody's indicates that downward rating pressure could arise on KEDLI's rating if the metric was expected to fall persistently below 15 percent. Therefore, unless the metric were expected to improve above 15 percent in future years, KEDLI would be at risk of a downgrade from Baa1 to Baa2. KEDNY's metric is below 15 percent but above 13 percent and, therefore, consistent with its current Baa2 rating. As indicated in the latest Moody's report, KEDNY's rating could be upgraded if the metric were at least 15 percent on a sustainable business and state climate measures did not lead to a deterioration in KEDNY's business risk profile. While the rating agencies normally focus on a multi-year trend rather than an individual year, to mitigate the risk that an unexpected shock leads to a downgrade, we believe it is prudent to target a metric in any individual year that provides ample headroom above the level at which the rating agencies indicate a downgrade could occur. The issue of headroom above metric thresholds has been recently highlighted as an industry risk by the rating agencies. For example, in its report "The Outlook for North American Regulated Utilities Turns Stable" from May 2023, S&P indicate that a significant risk for the industry is "the practice of many companies to operate with minimal financial cushion from their downgrade thresholds."

Q. What actions could be taken to increase cash flow to ensure that KEDLI's CFO pre-WC/debt metric remained above 15 percent in the Rate Year to safeguard its Baa1 Moody's credit rating?

A. There are multiple options to improve cash flow, for example a higher ROE or a capital structure with a higher equity ratio. In addition, KEDLI's adjusted CFO pre-WC is negatively impacted by the amortization of KEDLI's net regulatory liability proposed by the Staff Revenue Requirements Panel. We oppose this proposal as it is a contributing factor to KEDLI's CFO pre-WC/debt metric being below 15 percent and therefore could put the Company's Baa1 rating at risk. Removing this amortization would increase cash flow by $8.214 million in the Rate Year, resulting in a 33 basis point improvement in the metric to 14.98 percent. This adjustment, however, would not be enough to produce a metric above 15 percent. Moreover, as we noted, the rating agencies would need to see "at least 19% on a sustainable basis under its next rate plan" for an upgrade and that "downward rating pressure could arise" if the metric "was expected to fall persistently below 15%, or if state energy policy resulted in a deterioration in either cash flow predictability; or ability or timeliness of cost recovery."

Q. Is there anything else in Staff's proposals that could be viewed negatively by credit rating agencies or negatively impact the Companies' credit metrics?

A. Yes. In the testimony of the Staff Site Investigation and Remediation ("SIR") Panel, Staff reflects a downward adjustment of $21.041 million to KEDNY's forecast SIR expense. If the actual Rate Year expense is as per the Company's forecast, Staff's proposals would lower the Moody's adjusted CFO pre-WC/debt metric for the Rate Year by 51 basis points.

In its latest March 2023 rating report, Moody's cites KEDNY's "material and growing spending on environmental remediation" as a credit challenge. In addition, we note that Staff is proposing significant increases to the customer service quality penalties at risk (Staff Consumer Services Panel ("SCSP") at 73-77) and does not support any of the Companies' proposed Earning Adjustment Mechanisms (Staff Efficiency Panel ("SEP") at 61, 68-69, 75-76, and 81-84). The ability to recover costs in a timely fashion is important for the rating agencies and increased penalties could weaken their assessment of the Companies. For example, within the Moody's rating methodology for the Companies (Exhibit ___(KXD-14)), "Timeliness of Recovery of Operating and Capital Costs" has a sub-factor weighting of 12.5 percent in determining the scorecard indicated rating outcome.

VI. Service Company Asset Recovery Charge Rate of Return

Q. Does Staff Witness Duah agree with the Companies' proposed rate of return for the Service Company asset recovery charge?

A. No, he does not. Although he agrees with the Company's proposed debt cost rate, he substitutes his recommended ROE of 9.1 percent and a 48 percent equity ratio into the rate of return calculation. In support of his capital structure adjustment, Staff Witness Duah cites to several New York rate settlements where the equity ratio used for the Service Company asset recovery charge was aligned with the revenue requirement equity ratio, but concedes that the settlement in the 2019 KEDNY and KEDLI Rate Cases used the Service Company capital structure proposed by the Companies in this case.

Q. Do you agree with Staff Witness Duah's proposal?

A. Not entirely. We agree on the Service Company's debt cost rate. We also agree that the ROE authorized for the Companies should be used to determine the cost of equity component of the Service Company's rate of return. However, we continue to believe that the 50 percent equity ratio we proposed for the Service Company is appropriate. First, because it is the Service Company's actual equity ratio, which is separate and distinct from the Companies' equity ratios, and second, because the Service Company is not a public utility and there is no reason to assign the same capital structure to it as to a regulated utility. Regarding the ROE, as discussed in Company Witness Nowak's rebuttal testimony, 9.10 percent is an unreasonable estimate of the Companies' cost of equity. As such, Staff Witness Duah's proposed pre-tax rate of return of 6.89 percent for the Service Company, which is based on a 9.10 percent ROE and 48 percent equity ratio, is also unreasonable. Therefore, we continue to support the Companies' 7.32 percent pre-tax rate of return for the Service Company.

Q. Does this conclude your testimony?

A. Yes.

List of Exhibits

Exhibit___(CSP-1R)	Schedule 1: Updated Cost of Long-Term Debt
	Schedule 2: Updated Weighted Cost of Capital for the Rate Year and Data Years

Exhibit___(CSP-1R)

Schedule 1: Updated Cost of Long-Term Debt

Schedule 2: Updated Weighted Cost of Capital for the Rate Year and Data Years

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
WEIGHTED AVERAGE COST OF LONG-TERM DEBT
For 12 months period Ending December 31, 2022
($000)

LONG-TERM DEBT	RATE %	ISSUANCE DATE	MATURITY DATE	FACE VALUE	AVERAGE AMOUNT OUTSTANDING	ANNUAL INTEREST & FEES	ANNUAL AMORTIZATION DEBT DISCOUNT AND EXPENSE	TOTAL INTEREST AND ANNUAL AMORTIZATION	EFFECTIVE RATE
Senior Unsecured Debt due 2026	3.407%	3/10/2016	3/10/2026	500,000	500,000	17,035	210	17,245	3.45%
Senior Unsecured Debt due 2046	4.504%	3/10/2016	3/10/2046	500,000	500,000	22,520	128	22,648	4.53%
Senior Unsecured Debt due 2048	4.273%	3/14/2018	3/15/2048	650,000	650,000	27,775	184	27,959	4.30%
Senior Unsecured Debt due 2029	3.865%	3/4/2019	3/4/2029	550,000	550,000	21,258	235	21,493	3.91%
Senior Unsecured Debt due 2049	4.487%	3/4/2019	3/4/2049	450,000	450,000	20,192	125	20,317	4.51%
Amortization of expenses of repurchased bonds [1]	-	-	-	-	-	-	490	490	-
				2,650,000	**2,650,000**	**108,779**	**1,372**	**110,151**	**4.16%**
Repayments and Issuances									
Prorated Repayments:									
Term Loan	Variable	9/28/2021	8/11/2022	400,000	243,288	3,420	108	3,528	1.45%
Prorated New Issuances:									
Senior Unsecured Debt due 2027	4.632%	8/5/2022	8/5/2027	400,000	162,192	7,514	92	7,606	4.69%
Senior Unsecured Debt due 2032	4.866%	8/5/2022	8/5/2032	400,000	162,192	7,894	71	7,965	4.91%
Total Long-Term Debt				**3,850,000**	**3,217,671**	**127,606**	**1,643**	**129,250**	**4.02%**
Debt Outstanding at End of Year				**3,450,000**					

Notes
[1] Amortization of expenses of repurchased bonds relates to the buybacks of debt in 2016 and 2018, as per the Company's filings in case 15-G-0309.

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
WEIGHTED AVERAGE COST OF LONG-TERM DEBT
For 12 months period Ending March 31, 2025
($000)

LONG-TERM DEBT	RATE %	ISSUANCE DATE	MATURITY DATE	FACE VALUE	AVERAGE AMOUNT OUTSTANDING	ANNUAL INTEREST & FEES	ANNUAL AMORTIZATION DEBT DISCOUNT AND EXPENSE	TOTAL INTEREST AND ANNUAL AMORTIZATION	EFFECTIVE RATE
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Senior Unsecured Debt due 2049	4.487%	3/4/2019	3/4/2049	450,000	450,000	20,192	125	20,317	4.51%
Senior Unsecured Debt due 2027	4.632%	8/5/2022	8/5/2027	400,000	400,000	18,528	226	18,754	4.69%
Senior Unsecured Debt due 2032	4.866%	8/5/2022	8/5/2032	400,000	400,000	19,464	175	19,639	4.91%
Amortization of expenses of repurchased bonds [1]	-	-	-	-	-	-	268	268	-
				3,450,000	**3,450,000**	**146,771**	**1,551**	**148,321**	**4.30%**
Repayments and Issuances									
Prorated Repayments:									
Prorated New Issuances: [2]/[3]									
Senior Unsecured Debt due 2033	6.388%	9/15/2023	9/15/2033	400,000	400,000	25,552	206	25,758	6.44%
Senior Unsecured Debt due 2054	6.556%	6/30/2024	6/30/2054	500,000	375,342	24,608	103	24,711	6.58%
Total Long-Term Debt				**4,350,000**	**4,225,342**	**196,930**	**1,859**	**198,790**	**4.70%**
Debt Outstanding at End of Year				**4,350,000**					

Notes
[1] Amortization of expenses of repurchased bonds relates to the buybacks of debt in 2016 and 2018, as per the Company's filings in case 15-G-0309.
[2] New issuance rate for June 2024 issuance has been estimated using US Treasury forward rates as of September 12, 2023 plus credit spreads of 2.120% for 10-year issuances and 2.383% for 30-year issuances.
[3] Total expenses for the bond issued on September 15, 2023 have not yet been finalised and so annual amortization is based on an estimate.

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
WEIGHTED AVERAGE COST OF LONG-TERM DEBT
For 12 months period Ending March 31, 2026
($000)

LONG-TERM DEBT	RATE %	ISSUANCE DATE	MATURITY DATE	FACE VALUE	AVERAGE AMOUNT OUTSTANDING	ANNUAL INTEREST & FEES	ANNUAL AMORTIZATION DEBT DISCOUNT AND EXPENSE	TOTAL INTEREST AND ANNUAL AMORTIZATION	EFFECTIVE RATE
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Senior Unsecured Debt due 2033	6.388%	9/15/2023	9/15/2033	400,000	400,000	25,552	206	25,758	6.44%
Senior Unsecured Debt due 2054	6.556%	6/30/2024	6/30/2054	500,000	500,000	32,781	137	32,918	6.58%
Amortization of expenses of repurchased bonds [1]	-	-	-	-	-	-	142	142	-
				3,850,000	**3,850,000**	**188,068**	**1,558**	**189,626**	**4.93%**
Repayments and Issuances									
Prorated Repayments:									
Senior Unsecured Debt due 2026	3.407%	3/10/2016	3/10/2026	500,000	469,863	16,008	197	16,206	3.45%
Prorated New Issuances: [2]									
Senior Unsecured Debt due 2035	6.310%	6/30/2025	6/30/2035	400,000	300,274	18,946	148	19,094	6.36%
Senior Unsecured Debt due 2056	6.507%	2/28/2026	2/28/2056	400,000	33,973	2,211	10	2,220	6.54%
Total Long-Term Debt				**5,150,000**	**4,654,110**	**225,233**	**1,913**	**227,146**	**4.88%**
Debt Outstanding at End of Year				**4,650,000**					

Notes
[1] Amortization of expenses of repurchased bonds relates to the buybacks of debt in 2016 and 2018, as per the Company's filings in case 15-G-0309.
[2] New issuance rates have been estimated using US Treasury forward rates as of September 12, 2023 plus credit spreads of 2.120% for 10-year issuances and 2.383% for 30-year issuances.

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
WEIGHTED AVERAGE COST OF LONG-TERM DEBT
For 12 months period Ending March 31, 2027
($000)

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Senior Unsecured Debt due 2032	4.866%	8/5/2022	8/5/2032	400,000	400,000	19,464	175	19,639	4.91%
Senior Unsecured Debt due 2033	6.388%	9/15/2023	9/15/2033	400,000	400,000	25,552	206	25,758	6.44%
Senior Unsecured Debt due 2054	6.556%	6/30/2024	6/30/2054	500,000	500,000	32,781	137	32,918	6.58%
Senior Unsecured Debt due 2035	6.310%	6/30/2025	6/30/2035	400,000	400,000	25,239	197	25,436	6.36%
Senior Unsecured Debt due 2056	6.507%	2/28/2026	2/28/2056	400,000	400,000	26,030	112	26,142	6.54%
				4,650,000	**4,650,000**	**239,336**	**1,726**	**241,062**	**5.18%**
Repayments and Issuances									
Prorated Repayments:									
Prorated New Issuances: [1]									
Senior Unsecured Debt due 2036	6.394%	7/31/2026	7/31/2036	500,000	332,877	21,284	158	21,442	6.44%
Total Long-Term Debt	NOTE 1			**5,150,000**	**4,982,877**	**260,620**	**1,883**	**262,504**	**5.27%**
Debt Outstanding at End of Year				**5,150,000**					

Notes
[1] New issuance rates have been estimated using US Treasury forward rates as of September 12, 2023 plus credit spreads of 2.120% for 10-year issuances and 2.383% for 30-year issuances.

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
WEIGHTED AVERAGE COST OF LONG-TERM DEBT
For 12 months period Ending March 31, 2028
($000)

LONG-TERM DEBT	RATE %	ISSUANCE DATE	MATURITY DATE	FACE VALUE	AVERAGE AMOUNT OUTSTANDING	ANNUAL INTEREST & FEES	ANNUAL AMORTIZATION DEBT DISCOUNT AND EXPENSE	TOTAL INTEREST AND ANNUAL AMORTIZATION	EFFECTIVE RATE
Senior Unsecured Debt due 2046	4.504%	3/10/2016	3/10/2046	500,000	500,000	22,520	128	22,648	4.53%
Senior Unsecured Debt due 2048	4.273%	3/14/2018	3/15/2048	650,000	650,000	27,775	184	27,959	4.30%
Senior Unsecured Debt due 2029	3.865%	3/4/2019	3/4/2029	550,000	550,000	21,258	235	21,493	3.91%
Senior Unsecured Debt due 2049	4.487%	3/4/2019	3/4/2049	450,000	450,000	20,192	125	20,317	4.51%
Senior Unsecured Debt due 2032	4.866%	8/5/2022	8/5/2032	400,000	400,000	19,464	175	19,639	4.91%
Senior Unsecured Debt due 2033	6.388%	9/15/2023	9/15/2033	400,000	400,000	25,552	206	25,758	6.44%
Senior Unsecured Debt due 2054	6.556%	6/30/2024	6/30/2054	500,000	500,000	32,781	137	32,918	6.58%
Senior Unsecured Debt due 2035	6.310%	6/30/2025	6/30/2035	400,000	400,000	25,239	197	25,436	6.36%
Senior Unsecured Debt due 2056	6.507%	2/28/2026	2/28/2056	400,000	400,000	26,030	112	26,142	6.54%
Senior Unsecured Debt due 2036	6.394%	7/31/2026	7/31/2036	500,000	500,000	31,970	237	32,207	6.44%
				4,750,000	**4,750,000**	**252,778**	**1,737**	**254,515**	**5.36%**
Repayments and Issuances									
Prorated Repayments:									
Senior Unsecured Debt due 2027	4.632%	8/5/2022	8/5/2027	400,000	138,082	6,396	78	6,474	4.69%
Prorated New Issuances: [1]									
Senior Unsecured Debt due 2037	6.488%	7/31/2027	7/31/2037	550,000	367,671	23,853	172	24,025	6.53%
Total Long-Term Debt				**5,700,000**	**5,255,753**	**283,027**	**1,987**	**285,014**	**5.42%**
Debt Outstanding at End of Year				**5,300,000**					

Notes
[1] New issuance rates have been estimated using US Treasury forward rates as of September 12, 2023 plus credit spreads of 2.120% for 10-year issuances and 2.383% for 30-year issuances.

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One-Year Rate Plan

	Capitalization Ratio	Cost Rate	Weighted Cost
Long-Term Debt	51.82%	4.70%	2.44%
Customer Deposits	0.18%	3.45%	0.01%
Common Equity	48.00%	9.80%	4.70%
Total	100.00%		7.15%

Four-Year Rate Plan (ROE includes stay-out premium, forecasted cost of debt)

	Rate Year (FY25)			Data Year 1 (FY26)			Data Year 2 (FY27)			Data Year 3 (FY28)		
	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost
Long-Term Debt	51.82%	4.70%	2.44%	51.84%	4.88%	2.53%	51.85%	5.27%	2.73%	51.85%	5.42%	2.81%
Customer Deposits	0.18%	3.45%	0.01%	0.16%	3.45%	0.01%	0.15%	3.45%	0.01%	0.15%	3.45%	0.01%
Common Equity	48.00%	10.30%	4.94%	48.00%	10.30%	4.94%	48.00%	10.30%	4.94%	48.00%	10.30%	4.94%
Total	100.00%		7.39%	100.00%		7.48%	100.00%		7.68%	100.00%		7.76%

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
WEIGHTED AVERAGE COST OF LONG-TERM DEBT
For 12 months period Ending March 31, 2025
($000)

LONG-TERM DEBT	RATE %	ISSUANCE DATE	MATURITY DATE	FACE VALUE	AVERAGE AMOUNT OUTSTANDING	ANNUAL INTEREST & FEES	ANNUAL AMORTIZATION DEBT DISCOUNT AND EXPENSE	TOTAL INTEREST AND ANNUAL AMORTIZATION	EFFECTIVE RATE
Senior Unsecured Debt due 2041	5.819%	3/28/2011	4/1/2041	500,000	500,000	29,095	113	29,208	5.84%
Senior Unsecured Debt due 2026	2.742%	8/2/2016	8/15/2026	700,000	700,000	19,194	292	19,486	2.78%
Senior Unsecured Debt due 2052	3.586%	1/18/2022	1/18/2052	400,000	400,000	14,344	107	14,451	3.61%
				1,600,000	**1,600,000**	**62,633**	**512**	**63,145**	**3.95%**
Repayments and Issuances									
Prorated Repayments:									
Prorated New Issuances: [1]									
Senior Unsecured Debt due 2033	5.994%	3/6/2023	3/6/2033	500,000	500,000	29,970	237	30,207	6.04%
Senior Unsecured Debt due 2054	6.445%	11/30/2024	11/30/2054	400,000	132,603	8,546	37	8,583	6.47%
Total Long-Term Debt				**2,500,000**	**2,232,603**	**101,149**	**786**	**101,935**	**4.57%**
Debt Outstanding at End of Year				**2,500,000**					

Notes
[1] New issuance rates have been estimated using US Treasury forward rates as of September 12, 2023 plus credit spreads of 2.039% for 10-year issuances and 2.289% for 30-year issuances.

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
WEIGHTED AVERAGE COST OF LONG-TERM DEBT
For 12 months period Ending March 31, 2026
($000)

LONG-TERM DEBT	RATE %	ISSUANCE DATE	MATURITY DATE	FACE VALUE	AVERAGE AMOUNT OUTSTANDING	ANNUAL INTEREST & FEES	ANNUAL AMORTIZATION DEBT DISCOUNT AND EXPENSE	TOTAL INTEREST AND ANNUAL AMORTIZATION	EFFECTIVE RATE
Senior Unsecured Debt due 2041	5.819%	3/28/2011	4/1/2041	500,000	500,000	29,095	113	29,208	5.84%
Senior Unsecured Debt due 2026	2.742%	8/2/2016	8/15/2026	700,000	700,000	19,194	292	19,486	2.78%
Senior Unsecured Debt due 2052	3.586%	1/18/2022	1/18/2052	400,000	400,000	14,344	107	14,451	3.61%
Senior Unsecured Debt due 2033	5.994%	3/6/2023	3/6/2033	500,000	500,000	29,970	237	30,207	6.04%
Senior Unsecured Debt due 2054	6.445%	11/30/2024	11/30/2054	400,000	400,000	25,778	112	25,890	6.47%
				2,500,000	**2,500,000**	**118,381**	**861**	**119,242**	**4.77%**
Repayments and Issuances									
Prorated Repayments:									
Prorated New Issuances: [1]									
Senior Unsecured Debt due 2035	6.234%	8/31/2025	8/31/2035	450,000	261,370	16,294	126	16,420	6.28%
Total Long-Term Debt				**2,950,000**	**2,761,370**	**134,675**	**988**	**135,663**	**4.91%**
Debt Outstanding at End of Year				**2,950,000**					

Notes
[1] New issuance rates have been estimated using US Treasury forward rates as of September 12, 2023 plus credit spreads of 2.039% for 10-year issuances and 2.289% for 30-year issuances.

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
WEIGHTED AVERAGE COST OF LONG-TERM DEBT
For 12 months period Ending March 31, 2027
($000)

LONG-TERM DEBT	RATE %	ISSUANCE DATE	MATURITY DATE	FACE VALUE	AVERAGE AMOUNT OUTSTANDING	ANNUAL INTEREST & FEES	ANNUAL AMORTIZATION DEBT DISCOUNT AND EXPENSE	TOTAL INTEREST AND ANNUAL AMORTIZATION	EFFECTIVE RATE
Senior Unsecured Debt due 2041	5.819%	3/28/2011	4/1/2041	500,000	500,000	29,095	113	29,208	5.84%
Senior Unsecured Debt due 2052	3.586%	1/18/2022	1/18/2052	400,000	400,000	14,344	107	14,451	3.61%
Senior Unsecured Debt due 2033	5.994%	3/6/2023	3/6/2033	500,000	500,000	29,970	237	30,207	6.04%
Senior Unsecured Debt due 2054	6.445%	11/30/2024	11/30/2054	400,000	400,000	25,778	112	25,890	6.47%
Senior Unsecured Debt due 2035	6.234%	8/31/2025	8/31/2035	450,000	450,000	28,054	217	28,271	6.28%
				2,250,000	**2,250,000**	**127,241**	**787**	**128,027**	**5.69%**
Repayments and Issuances									
Prorated Repayments:									
Senior Unsecured Debt due 2026	2.742%	8/2/2016	8/15/2026	700,000	260,822	7,152	109	7,261	2.78%
Prorated New Issuances: [1]									
Senior Unsecured Debt due 2036	6.320%	8/31/2026	8/31/2036	450,000	261,370	16,519	126	16,645	6.37%
Senior Unsecured Debt due 2056	6.406%	8/31/2026	8/31/2056	450,000	261,370	16,744	73	16,816	6.43%
Total Long-Term Debt				**3,850,000**	**3,033,562**	**167,655**	**1,094**	**168,749**	**5.56%**
Debt Outstanding at End of Year				**3,150,000**					

Notes
[1] New issuance rates have been estimated using US Treasury forward rates as of September 12, 2023 plus credit spreads of 2.039% for 10-year issuances and 2.289% for 30-year issuances.

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID
WEIGHTED AVERAGE COST OF LONG-TERM DEBT
For 12 months period Ending March 31, 2028
($000)

LONG-TERM DEBT	RATE %	ISSUANCE DATE	MATURITY DATE	FACE VALUE	AVERAGE AMOUNT OUTSTANDING	ANNUAL INTEREST & FEES	ANNUAL AMORTIZATION DEBT DISCOUNT AND EXPENSE	TOTAL INTEREST AND ANNUAL AMORTIZATION	EFFECTIVE RATE
Senior Unsecured Debt due 2041	5.819%	3/28/2011	4/1/2041	500,000	500,000	29,095	113	29,208	5.84%
Senior Unsecured Debt due 2052	3.586%	1/18/2022	1/18/2052	400,000	400,000	14,344	107	14,451	3.61%
Senior Unsecured Debt due 2033	5.994%	3/6/2023	3/6/2033	500,000	500,000	29,970	237	30,207	6.04%
Senior Unsecured Debt due 2054	6.445%	11/30/2024	11/30/2054	400,000	400,000	25,778	112	25,890	6.47%
Senior Unsecured Debt due 2035	6.234%	8/31/2025	8/31/2035	450,000	450,000	28,054	217	28,271	6.28%
Senior Unsecured Debt due 2036	6.320%	8/31/2026	8/31/2036	450,000	450,000	28,441	217	28,658	6.37%
Senior Unsecured Debt due 2056	6.406%	8/31/2026	8/31/2056	450,000	450,000	28,827	125	28,952	6.43%
				3,150,000	**3,150,000**	**184,509**	**1,129**	**185,638**	**5.89%**

Repayments and Issuances

Prorated Repayments:

Prorated New Issuances:

Total Long-Term Debt	**3,150,000**	**3,150,000**	**184,509**	**1,129**	**185,638**	**5.89%**
Debt Outstanding at End of Year	**3,150,000**					

Notes

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One-Year Rate Plan

	Capitalization Ratio	Cost Rate	Weighted Cost
Long-Term Debt	51.70%	4.57%	2.36%
Customer Deposits	0.30%	3.45%	0.01%
Common Equity	48.00%	9.80%	4.70%
Total	100.00%		7.07%

Four-Year Rate Plan (ROE includes stay-out premium, forecasted cost of debt)

	Rate Year (FY25)			Data Year 1 (FY26)			Data Year 2 (FY27)			Data Year 3 (FY28)		
	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost	Capitalization Ratio	Cost Rate	Weighted Cost
Long-Term Debt	51.70%	4.57%	2.36%	51.75%	4.91%	2.54%	51.77%	5.55%	2.87%	51.78%	5.89%	3.05%
Customer Deposits	0.30%	3.45%	0.01%	0.25%	3.45%	0.01%	0.23%	3.45%	0.01%	0.22%	3.45%	0.01%
Common Equity	48.00%	10.30%	4.94%	48.00%	10.30%	4.94%	48.00%	10.30%	4.94%	48.00%	10.30%	4.94%
Total	100.00%		7.31%	100.00%		7.49%	100.00%		7.82%	100.00%		8.00%